

Alvaro Castaneda

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Technical Artist (Software Engineering) with 20+ years of experience developing procedural workflows, scalable content pipelines, and automation tools for VFX, graphics, and AI applications. Specialized in large-scale asset systems, workflow design, and production infrastructure.

EXPERIENCE

Technical Artist (Software Engineering) — Meta Reality Labs

Aug. 2021 – Present

Remote, US

- Designed and built large-scale data processing pipelines handling 5M+ 3D assets from commercial and open-source providers to support generative AI model training workflows.
- Built AI-agent-powered tools integrated directly into Houdini, enabling natural language interaction, workflow automation, and scalable procedural content generation.
- Developed VLM-based automated QA and visual validation pipelines for generated assets at scale.
- Procedural hair systems contributed to published research: HairSapiens and Quaffure (real-time neural hair simulation).
- Served as primary Houdini platform authority across Reality Labs, supporting tooling, onboarding, and pipeline workflows.

Houdini Procedural Artist — AI.Reverie

Jan. 2019 – Dec. 2021

Greater New York City Area

- Houdini Procedural TD on RarePlanes Synthetic Dataset (In-Q-Tel / CosmiQ Works); built systems generating thousands of asset variations for ML and Computer Vision applications.
- Developed Python pipeline tools to manage thousands of assets; built Unreal Engine spawning and point-cloud instancing systems for large-scale city scenes.

Founder / VFX Artist — MIX Studio

Feb. 2012 – Dec. 2021

Ciudad Juárez, Mexico

- Developed procedural VFX workflows and simulation pipelines for film, TV, and advertising productions including Jumanji: Welcome to the Jungle.
- Founded a globally recognized Houdini training platform; produced content in partnership with SideFX, Redshift, Arnold, and Pixar RenderMan.

Teaching & Mentorship

2009 – 2022

- **CGMA | Computer Graphics Master Academy — Houdini Instructor**
 - Taught Houdini fundamentals, procedural workflows, and FX techniques to artists transitioning into technical roles.
- **CMIVFX — Mentor / Technology Specialist**
 - Created professional training content for Houdini, Python, and Nuke used by artists and studios worldwide.
- **MIX Training — Founder / Instructor**
 - Developed Houdini-focused technical training courses and in collaboration with industry software vendors.

PROJECTS & PUBLICATIONS

Featured contributor in 3D WORLD magazine (4 publications).

Procedural hair systems contributed to published research papers including HairSapiens and Quaffure (real-time neural hair simulation).

TECHNICAL SKILLS

Languages: Python, C++, VEX, C, GLSL, Bash, Odin

Tools & DCC: Houdini, Maya, Blender, Nuke, Unreal Engine, OpenUSD, Solaris

AI / Pipelines: Generative AI workflows, Synthetic Data, VLM-based QA, Procedural Generation, Automation

Platforms: Linux, macOS, Windows, Git

EDUCATION

IEU Universidad - Bachelor of Science, Artificial Intelligence

Expected 2026

Machine Learning Specialization - DeepLearning.AI, Stanford University

Udacity - C++ Developer Nanodegree, Computer Science